



IDP

Problem Definition

Following the requirements in our IDP competition:

- 1. Size Constraint: 18" x 18"
- 2. Time Constraint: t < 1 min
- 3. Mass Constraint of prototype: m < 20 lbs
- 4. Number of Beanbags launched: 3 beanbags at a time

We are tasked with designing a launching mechanism that will propel beanbag across 10ft, and into the cornhole on the other side of the track as shown in image below.



Figure 1: Contest Court Dimension

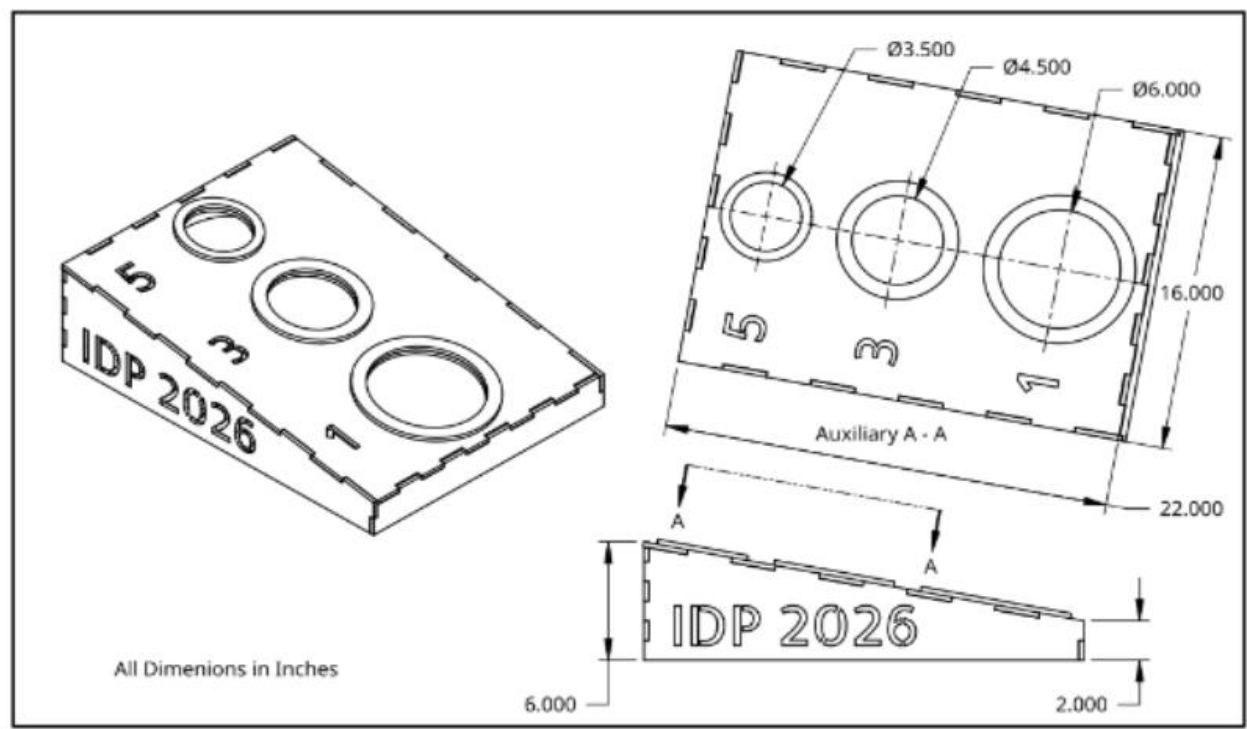
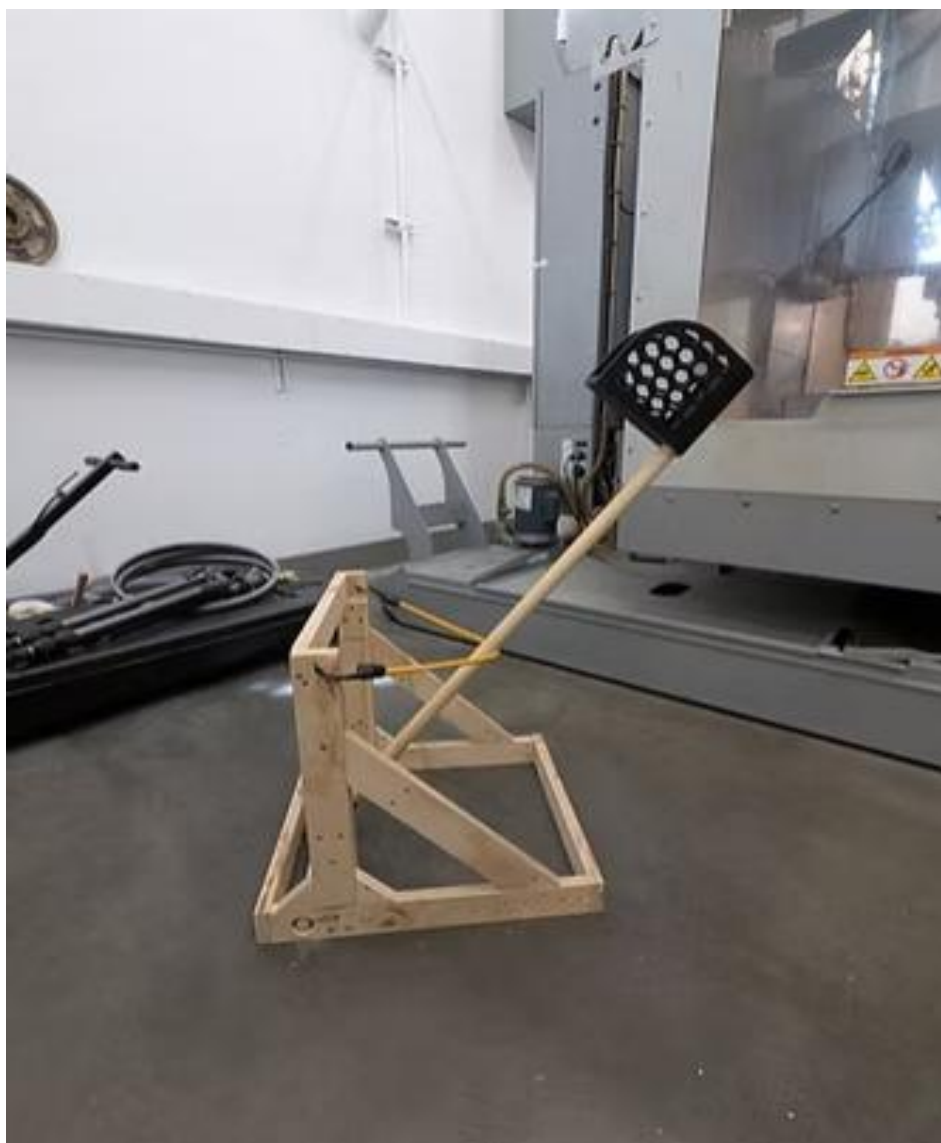


Figure 2: Target Dimension Details

Design Process

- Before Designing:
- conducted research on existing structures that serve similar purposes of launching projectiles
 - Designed the parts for the catapult collaboratively.
 - We collected materials needed and started to build a prototype of our design in parallel of our CAD design process, which was used to test the structure we designed and find any potential flaws.
 - Kept constant update of CAD design based on information gathered in prototype building.
 - Improvements were made based on data gathered from testing



Older version of catapult

Bean Bag Launcher Prototype

TEAM 201

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DESIGN SOLUTION

The Catapult we designed aims to launch 5x5-inch bean bags to cornholes 10 feet away with maximized accuracy and throwing speed while meeting the dimensions and weight requirements.

The base frame provides stable support to the arm, with counterweights to prevent the catapult from tripping over when the arm is pulled. We added Dasani water bottles on the forefront, and back of the frame, and in doing so we found our frame to be stable during and after launch. The catapult is designed to hold and launch 3 beanbags at a time, as the increase in the number of projectiles also increases the chance of projectiles hitting the cornholes.

We calibrated the catapult to aim for the 3-point hole, as it has a much higher reward compared to the 1-point hole, while it is easier to aim than the 5-point hole. The 3-point hole is also found in the middle among the 3 locations, which means any bean bags that flies too far or too close still have the chance to enter one of them if the deviation is not toward left or right.

Safety/Social Issues

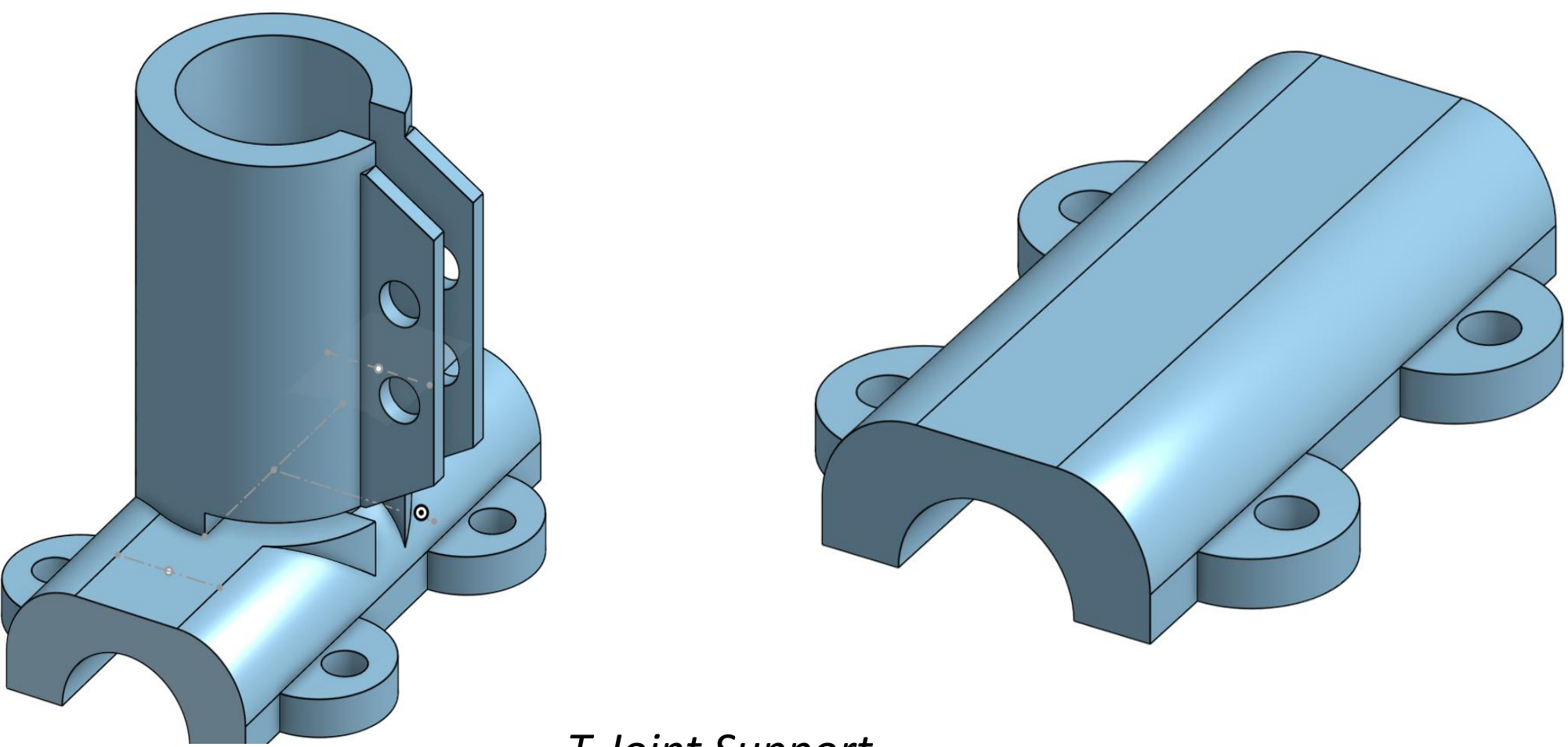
Our design is quite lightweight due to the use of wood and PLA in the design. This introduces a risk of injury during operation as the catapult can slip and the arm can impact the frame during fire. To mitigate this, we used Dasani water bottles along with shelf liner to keep the device secure.



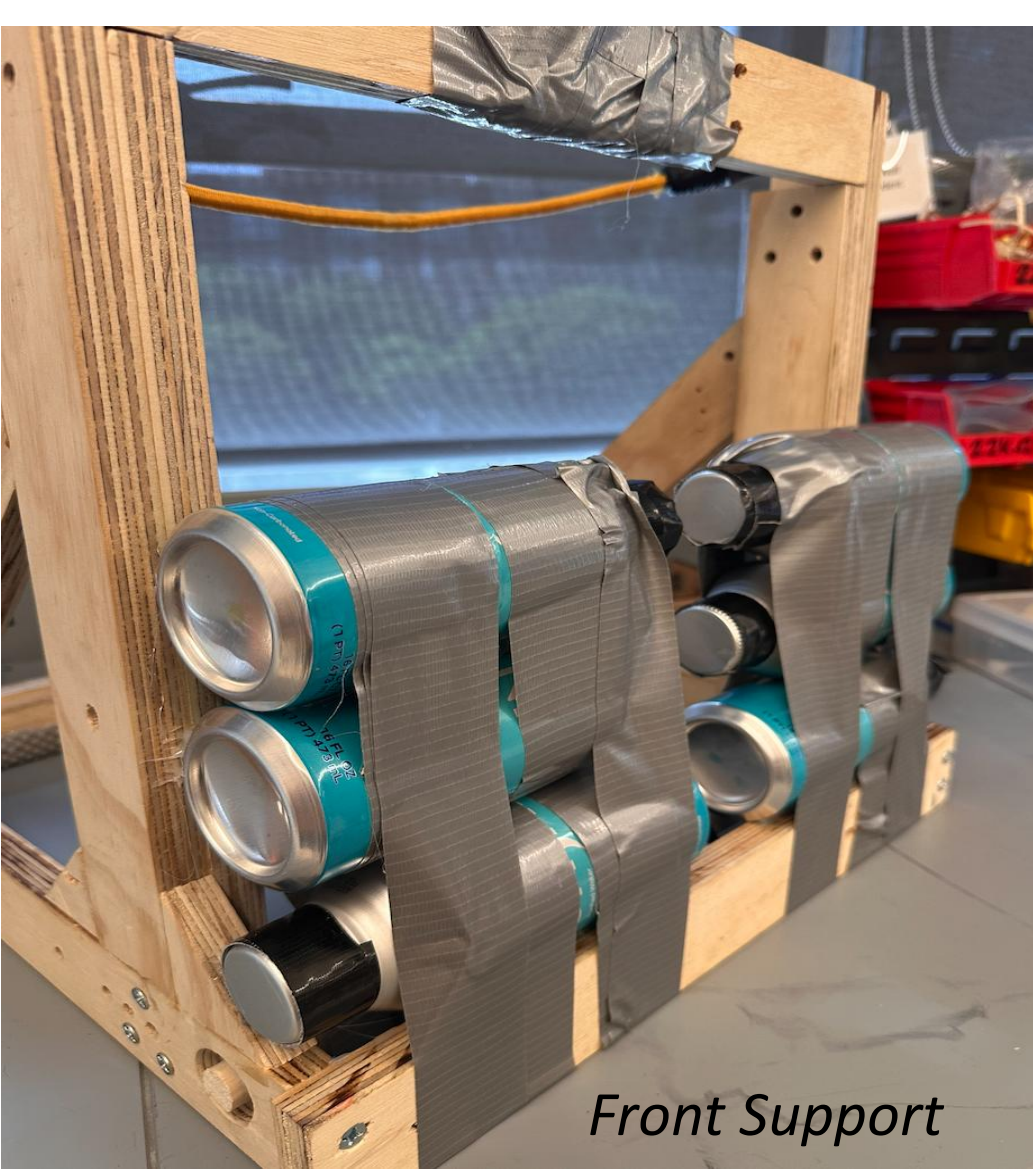
Dasani Water Bottle

T-joint Support

The T-Joint joins the frame axle and the catapult arm together. The top attaches to the bottom bracket with 10-24 screws to keep enough friction on the frame axle. The arm attaches to the joint in the same manner, with a cutout allowing the top part of the joint flex.



T-Joint Support



Front Support



Back Support

Bean Bag Holder Design

Our beanbag holder went through multiple design iterations (settled w/ v2) with primary goals being mass reduction, increased structural strength, and minimizing spread of the beanbags during and after launch.

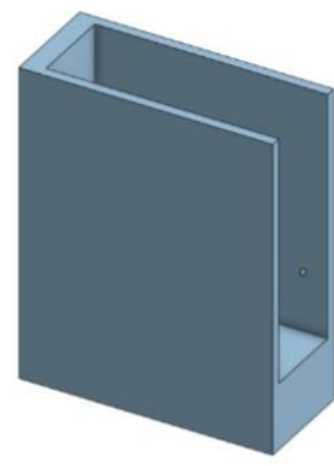


Figure 5: v1 Bean Bag Holder

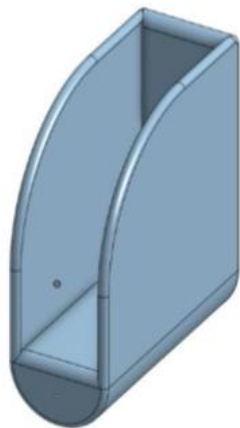


Figure 6: v2 Special Bean Bag Holder



Figure 6: v2: Special Bean Bag Holder (ignore the bean bag holder on frame design.)

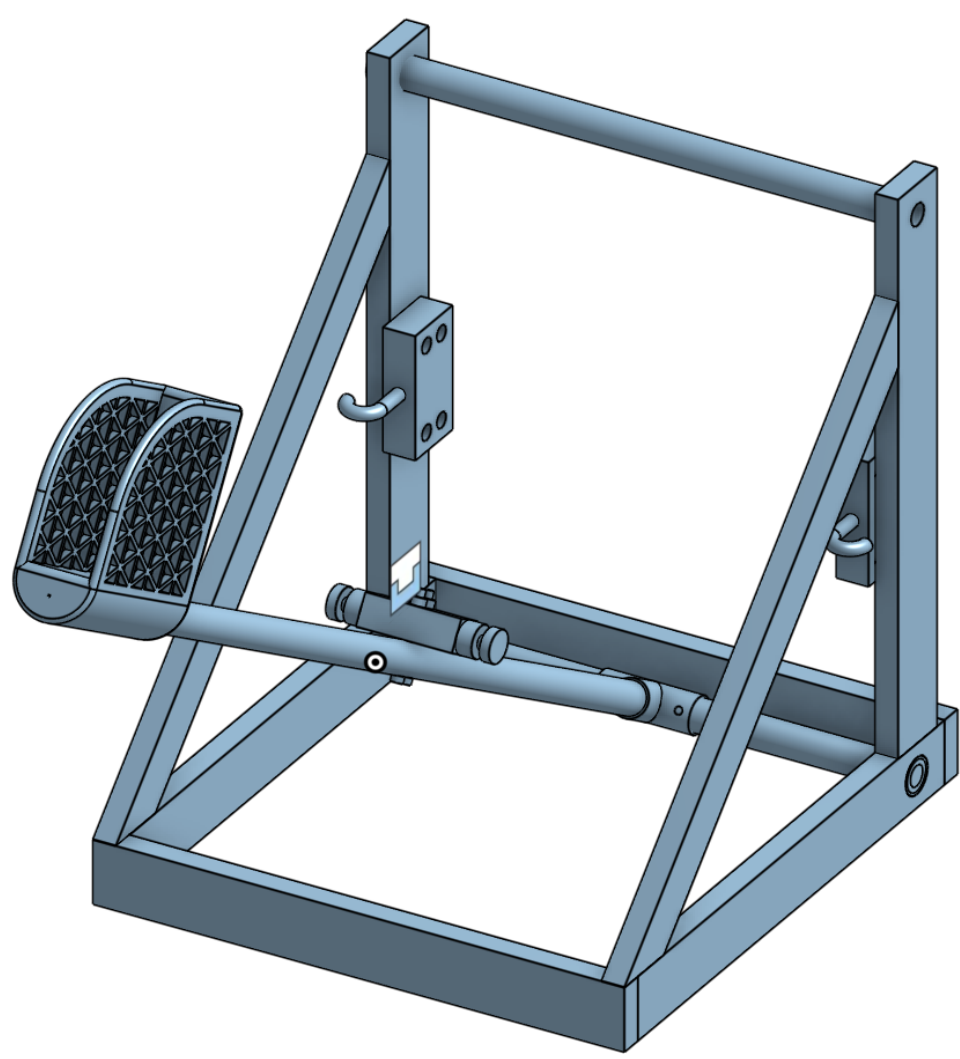
Cost Projections

	A	B	C	D	E
1	Items	Quantity	IDP Price	qty*price	
2	Plywood 1/8" (24"x18")	9	\$ 5.00	\$ 45.00	
3	Bearings	2	\$ 2.00	\$ 4.00	
4	Wood Dowel	2	\$ 0.75	\$ 1.50	
5	Nails/Bolts	25	\$ 0.05	\$ 1.25	
6	Bungee Cord	1	\$ 0.20	\$ 0.20	
7	PLA Filament (1.75 mm or 2.85mm)	278.42	\$ 0.03	\$ 8.35	grams
8	Total			\$ 60.30	



Market/Competition

Some competition to acknowledge would be the trebuchet and the crossbow. Where both are similar to the catapult where they throw the beanbags however they face some repercussion. For instance, the crossbow has the added factor of keeping into account friction. On the other hand, the trebuchet needs to have precise measurements to make sure the counter-weight along with the structure of the mechanical structure are not above the 20lb weight limit.



Initial CAD prototype



Final prototype

FUTURE WORK

- Improvements for the future:
- Earlier testing to have more time to find flaws
 - Increasing base stability by adding more surface area
 - Keeping arm joint strength/stability in mind to avoid arm joint from breaking
 - Keeping mass in mind when producing launcher arm to not risk being above weight limit
 - 3D print more parts for faster fabrication